Experiment -9

Python Program To Travelling Salesman Problem

Code:

```
from itertools import permutations

def travelling_salesman(graph, start):
    n = len(graph)
    cities = list(range(n))
    cities.remove(start)

    min_cost = float('inf')
    best_path = []

    for route in permutations(cities):
        path = [start] + list(route) + [start]
        cost = 0

        for i in range(len(path) - 1):
            cost += graph[path[i]][path[i + 1]]

        if cost < min_cost:
            min_cost = cost
            best_path = path

    return best_path, min_cost
```

```
graph = [
    [0, 10, 15, 20],
    [10, 0, 35, 25],
    [15, 35, 0, 30],
    [20, 25, 30, 0]
]

start = 0

path, cost = travelling_salesman(graph, start)

print("Shortest Path:", " -> ".join(map(str, path)))
print("Minimum Cost:", cost)
```

```python
from itertools import permutations

def travelling_salesman(graph, start):
    n = len(graph)
    cities = list(range(n))
    cities.remove(start)

    min_cost = float('inf')
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    for route in permutations(cities):
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        for i in range(len(path) - 1):
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graph = [
    [0, 10, 15, 20],
    [10, 0, 35, 25],
    [15, 35, 0, 30],
    [20, 25, 30, 0]
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start = 0

path, cost = travelling_salesman(graph, start)

print("Shortest Path:", " -> ".join(map(str, path)))
print("Minimum Cost:", cost)
```

```
Shortest Path: 0 -> 1 -> 3 -> 2 -> 0
Minimum Cost: 80


...Program finished with exit code 0
Press ENTER to exit console.
```